

Sebastian Rowan

ORISE POSTDOCTORAL RESEARCH FELLOW

USACE ERDC Coastal and Hydraulics Laboratory, Vicksburg, MS

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I am postdoctoral research fellow studying the impacts of flood events and climate change on people and communities. The goal of my research is to develop a more comprehensive understanding of the risks posed by floods to enable the development of mitigation efforts that prioritize long-term sustainability and community well-being.

🎓 Education

University of New Hampshire

Durham, New Hampshire

PH.D., CIVIL AND ENVIRONMENTAL ENGINEERING

September 2026

- Dissertation Title: Quantifying Greenhouse Gas Emissions and Mental Health Impacts as Co-Benefits of Flood Risk Mitigation
- Advisor: Dr. Weiwei Mo

University of New Hampshire

Durham, New Hampshire

B.S. ENVIRONMENTAL ENGINEERING

2016

🏢 Experience

U.S. Army Corps of Engineers, Engineer Research Development Center

Vicksburg, MS

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Sep. 2026 - Present

U.S. Army Corps of Engineers, Engineer Research Development Center

Vicksburg, MS

ORISE GRADUATE RESEARCH FELLOW

Sep. 2020 - Sep. 2026

- Contributed to the development of USACE'S "go-consequences" flood consequence analysis tool.
- Lead systematic literature review and meta-analysis to assess the effect of home damage on post-flood psychiatric morbidity to support USACE effort to comprehensively assess "Other Social Effects" of floods for planning studies
- Co-lead economic consequence and social vulnerability analysis of future flooding in the Mississippi River Valley
- Performed data analysis using social vulnerability and other geospatial data to support Tier 1 Other Social Effects/Environmental Justice Analysis for the USACE New York/New Jersey Harbors and Tributaries Coastal Storm Risk Management Feasibility Study

University of New Hampshire

Durham, NH

RESEARCH & TEACHING ASSISTANT

Aug. 2018 - Dec. 2020

- (RA) Resilient Bridge Planning in Mozambique - Bridge Failure Risk from Flooding and Climate Change
- (TA) CEE 705: Introduction to Sustainable Engineering (Fall 2019, Fall 2020)
- (TA) CEE 502: Project Engineering (Spring 2019)

New Hampshire Department of Transportation, Bureau of Planning and Community Assistance

Concord, NH

CIVIL ENGINEER I-II

2016 - 2018

University of New Hampshire InterOperability Laboratory

Durham, NH

10 GIGABIT ETHERNET TECHNICIAN

2014-2017

New Hampshire Department of Environmental Services, Air Resources Division

Concord, NH

ENVIRONMENTAL TECHNICIAN

2015

🏆 Awards

3RD PLACE - UNH 3-MINUTE THESIS (3MT) COMPETITION

2025

📄 Publications

Rowan, S., Yeates, E. (In Review). The effects of home damage on post-flood mental health: A systematic review and meta-analysis. *Natural Hazards Review*

Qiao, Y., Guo, Y., **Rowan, S.**, Medina, R., Espinet, X., Cullen, J., Meng, F., Cao, Z. (2026). Doubling of flood-induced bridge asset failure loss in Mozambique under 2050 climate. *Global Environmental Change*, <https://doi.org/10.1016/j.gloenvcha.2025.103103>

Rowan, S., Bell, E. S., Mo, W., (2025). Quantifying the hidden carbon cost of floods: a stochastic and uncertainty-based valuation framework. *Environmental Research Letters*, <https://doi.org/10.1088/1748-9326/ae2d76>

Rowan, S., Yeates, E., (2024) PROTOCOL: The effect of home damage on post-flood psychiatric morbidity: A systematic review and meta-analysis. PROSPERO, <https://www.crd.york.ac.uk/PROSPERO/view/CRD42024618891>

Memarsadeghi, N. P., **Rowan, S.**, Sisco, A. W., Tavakoly, A. A., (2024). Enhancing resilience: Integrating future flood modeling and socio-economic analysis in the face of climate change impacts. *Science of the Total Environment*, <https://doi.org/10.1016/j.scitotenv.2024.174893>

Seigerman, C. K., McKay, S. K., Basilio, R., Biesel, S. A., Hallemeier, J., Mansur, A. V., Piercy, C., **Rowan, S.**, Ubiali, B., Yeates, E., & Nelson, D. R. (2023). Operationalizing equity for integrated water resources management. *JAWRA Journal of the American Water Resources Association*, 59(2), 281–298. <https://doi.org/10.1111/1752-1688.13086>

Galaitis, S., Kurth, M., **Rowan, S.**, Yeates, E., & Kalaidjian, E. (2022). New York—New Jersey Harbor and Tributaries Coastal Storm Risk Management Feasibility Study—Tier 1 Other Social Effects/Environmental Justice Analysis. U.S. Army Corps of Engineers New York District. https://www.nan.usace.army.mil/Portals/37/Appendix%20A12_Tier%201%20SE_EJ_HATS.pdf

Rowan, S., & Kwiatkowski, K. (2020). Assessing the Relationship Between Social Vulnerability, Social Capital, and Housing Resilience. *Sustainability*, 12(18), 7718. <https://doi.org/10.3390/su12187718>

Presentations

Yeates, E., **Rowan, S.**, Bledsoe, B. Reconciling Flood Risk Management with Systemic Environmental Hazards. *AEES 2026 Annual Meeting; June 2026; Burlington, VT*. Poster.

Rowan, S., Han, F., Permenter, R., Mo, W. Monte Carlo-Based Building Damage and Emission Functions for Assessing Flood Risks. *AGU25; December 2025; New Orleans, LA*. Oral Presentation.

Rowan, S., Yeates, E. The Effect of Home Damage On Post-Flood Psychiatric Morbidity: A Systematic Review and Meta-Analysis. *AGU25; December 2025; New Orleans, LA*. Oral Presentation.

Rowan, S., Measuring what matters for flood risk management. *University of New Hampshire 3MT (3-Minute Thesis) Competition Finals; March 2025; Durham, NH*. Oral Presentation. https://media.unh.edu/media/Sebastian%20Rowan%20-%20UNH%203MT%20Finals%202025%20-%20Third%20Place/1_da4qt06x

Rowan, S., Memarsadeghi, N., Sisco, A., Tavakoly, A. An Assessment of the Socio-Economic Impacts from Climate Change and its Relationship with Vulnerability. *AGU23; December 2023; San Francisco, CA*. Oral Presentation.

Rowan, S., Yeates, E., Mo, W. Estimating the Greenhouse Gas Emissions of Flood Damages. *AEESP Research & Education Conference; June 2023; Boston, MA*. Poster.

Rowan, S., Yeates, E. Predicting the Mental Health Impacts of Floods. *47th Annual Natural Hazards Research and Applications Workshop; July 2022; Virtual*. Poster.

Rowan, S., Yeates, E., Wells, E. Quantifying the Health Impacts of Floods - A Systematic Literature Review. *2021 UNC Water and Health Conference; October 2021; Virtual*. Poster.

Rowan, S., Kwiatkowski, K. Assessing the Relationship Between Social Vulnerability, Social Capital, and Housing Resilience. *45th Annual Natural Hazards Research and Applications Workshop; July 2020; Virtual*. Poster.

Rowan, S., Kwiatkowski, K., Qiao, Y. Resilient Bridge Planning in Mozambique: Bridge Failure Risk from Flooding and Climate Change. *2nd International Conference on Transportation System Resilience to Natural Hazards and Extreme Weather Events (TR2019); November 2019; Washington, D.C*. Oral Presentation.

Open Source Projects

SVIBUILD R

Active

- An R package that allows users to download or construct the CDC's Social Vulnerability Index as a tidyverse or simple features data frame.
- Enables greater flexibility in region selection for SVI analyses than is possible with state- or national-level datasets hosted by CDC.

NSI DATA QGIS PLUGIN

Active

- A QGIS plugin that downloads data from the USACE National Structures Inventory for a specified region and adds it to a map.

Skills

PROGRAMMING LANGUAGES

- Python, R, Go, SQL, MATLAB

SOFTWARE

- QGIS, ArcGIS, go-consequences, Vensim, NVivo, Excel